Karanraj Duraikannu

Software Engineer
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Summary

Senior Software Engineer with 6 years of experience in architecting and delivering full-stack solutions. Proven ability to lead projects focused on application modernization (React to TypeScript conversion), performance optimization, and implementing robust test automation strategies. Combines technical expertise in Java and React with deep knowledge of DevOps, including containerization and orchestration, to enhance team productivity and ensure seamless, high-quality product delivery.

Technical Skills

- **Languages:** JavaScript (ES6), TypeScript, Python, HTML5, CSS3, Java
- **Frameworks/Libraries:** ReactJS, NextJs, Express.js, Node.js, jQuery, Webpack, Babel, Ant Design, Tailwind CSS
- **Databases:** MySQL, Postgres, MongoDB
- **DevOps/Tools:** Docker, Kubernetes, GitLab CI/CD, Git, VS Code, IntelliJ, PyCharm, Airflow, Nginx, Postman, JMeter, Selenium, Vitria
- **Methodologies:** Agile, RESTful APIs

Professional Experience

Freelance

SinceNew Technologies – Freelance

December 2025 – Present

Role: Full Stack Developer

Client: AVR Energies

Project: AI-Powered Talent Acquisition & Recruitment Portal

Project Description: Engineered a full-stack recruitment platform featuring an AI/ML-driven resume parser to automate candidate data extraction and streamline the "Easy Apply" process. Developed a comprehensive Admin & HR Dashboard to manage job postings, track applications, and perform bulk candidate data uploads for large-scale hiring. Built an automated email notification system to alert enrolled candidates of new openings, significantly increasing applicant engagement. The system enhances HR efficiency by centralizing candidate management and providing real-time analytics on job availability.

Technologies: Next.js, Express, MongoDB, Ant Design, Tailwind CSS.

- **Development:** Building a high-performance web platform for renewable energy services, utilizing **Next.js** for SEO-friendly rendering and **Tailwind CSS** for responsive UI.
- **Backend:** Architected a scalable REST API using **Express** and **MongoDB** to manage dynamic project data and user inquiries.
- **AI/ML Integration:** Automated data entry by extracting key details from uploaded resumes.
- **Bulk Processing:** Developed tools for HR to upload and manage high-volume candidate datasets via Excel/CSV integration.
- **Automated Communication:** Integrated SMTP/Email services for automated user engagement and status updates.

Tech Mahindra — Senior Software Engineer

September 2024–November 2025, Bangalore

Role: Full Stack Automation Developer

Client: Dish 5G wireless network

Project: Network Configuration & Monitoring Automation

Project Description: Architected a full-stack automation platform at Tech Mahindra to manage and monitor large-scale router configurations via Camunda BPA. Designed a React frontend for bulk configuration updates, significantly reducing manual deployment errors across the network. Integrated Vitria for continuous real-time log monitoring and built custom Python scripts for automated incident detection.

Engineered a seamless ServiceNow integration to trigger instant ticketing and team notifications upon detecting router failures.

Technologies: React, Camunda (BPA), Python, Vitria, ServiceNow API, REST APIs

- **Automated Network Orchestration:** Engineered a **Business Process Automation (BPA)** workflow using **Camunda** to automate complex configuration updates for network routers, reducing manual configuration errors and deployment time.
- **User-Centric Configuration Interface:** Developed a **React** front-end application that allows administrators to input and validate router data before pushing updates to the automated pipeline.
- **Proactive Incident Monitoring:** Integrated the **Vitria** monitoring tool with custom automation scripts to perform real-time log analysis and identify system failures.
- **Automated Incident Management:** Streamlined support workflows by programmatically generating **ServiceNow** tickets upon failure detection, ensuring immediate team notification and faster MTTR (Mean Time To Repair).
- Utilized the **React.js** library to create a dynamic UI with form functionality, enabling data submission and retrieval through REST API calls.
- Designed and implemented interactive forms using React.js, handling all API-based communication for data submission and retrieval.
- Developed a Rapid Infrastructure Automation (RIA) application with a user-friendly workflow to streamline and automate bare-metal provisioning and OS deployment through the MAAS API.
- Managed the end-to-end incident lifecycle from proactive detection in Vitria logs to resolution and closure, acting as a key operations team member.
- Automated critical network management tasks using **Business Process Automation (BPA)** tools, standardizing workflows for onboarding and decommissioning routers.

Orion Data Governance — Product Engineer

August 2019–September 2024, Bangalore

Role: UI Developer & Backend Integration Engineer

Client: Orion Data Governance

Project: Data Governance Project

Project Description: Developed a high-performance data governance tool that extracts metadata from diverse ecosystems including SQL, Redshift, and Terraform. Engineered a React-based UI to visualize end-to-end data lineage through complex graph structures for improved data transparency.

Technologies: React, Python, Java8, Camunda BPM, Selenium, Airflow, REST APIs, Jira, Slack, Git, Redux, Material UI

- Led the development and optimization of large-scale enterprise applications using **React, JavaScript, TypeScript, Node.js, and Java**, resulting in improved user experience and performance.
- Spearheaded a project to refactor existing ReactJS applications from Javascript to TypeScript, significantly improving code quality and long-term maintainability.
- Developed scalable, RESTful APIs using the **Spark Java** framework with integrated authentication validation to support microservices architecture.
- Engineered a core **Java** microservice to integrate with multiple external platforms, including Slack, Microsoft Teams, Jira, and ServiceNow.
- Automated UI regression testing using **Selenium** with **Java and Python**, reducing manual effort and preventing human errors.
- Built and integrated automated API testing with **Postman** and **Airflow** to validate API responses and performance, accelerating product delivery.
- Implemented Docker for containerizing services such as databases, UI/Nginx, and middleware, streamlining development and deployment processes.
- Performed load testing with **JMeter** to identify system bottlenecks, ensuring application stability and optimal performance under heavy traffic.
- Improved test coverage and accelerated bug detection by designing modular and reusable test scripts, incorporating design patterns like the Page Object Model.
- Collaborated with a distributed team using **GitLab** for version control and **CI/CD**, promoting modern development practices.
- Integrated automated testing into **CI/CD** pipelines to ensure faster, on-time product delivery and to minimize human error.
- **Workflow Orchestration & API Development:** Engineered a robust backend using **Java Spark** to interface with the **Camunda BPM engine**, automating complex administrative approval cycles.

- **Task Management Lifecycle:** Developed APIs to programmatically deploy BPMN models, retrieve real-time user tasks, and handle task assignments, approvals, and rejections.
- **Automated Communication Loop:** Integrated a dynamic email notification system triggered by process milestones to alert specific user roles of pending actions and status updates.
- **Data Pipeline Orchestration:** Developed **Apache Airflow** DAGs in **Python** to automate and monitor data extraction from input files, ensuring data integrity for governance processes

Education

Bachelor of Engineering in Computer Science
SVS College of Engineering, Coimbatore.
Graduated 2019

Personal Details

Date of Birth: 31 July 1997

Language: Tamil, English

Nationality: Indian

Marital status: Married

Native: Madurai, Tamilnadu.